



UNSW
SYDNEY

MATH2601 – Higher Linear Algebra

Topic 7 – Canonical forms

Lecture 7.1 – Generalised eigenspaces and Jordan normal form

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Generalised eigenspaces

Definition. Given a vector space $(V, \mathbb{C})$ with $T \in \mathcal{L}(V, V)$, for any eigenvalue λ of T , the **generalised eigenspace** of λ , written G_λ or $G_\lambda(T)$, is

$$G_\lambda = \{\mathbf{v} \in V : (T - \lambda \text{id})^k(\mathbf{v}) = \mathbf{0} \text{ for some } k \in \mathbb{Z}^+\}.$$

If $\mathbf{v} \in G_\lambda$ then $\mathbf{v}$ is called a **generalised λ -eigenvector** for T .

Notation. For any $k \in \mathbb{Z}^+$, we shall write $E_\lambda^k = \ker((T - \lambda \text{id})^k)$.

Clearly $E_\lambda = E_\lambda^1 \leq E_\lambda^2 \leq E_\lambda^3 \leq \dots$, since if $(T - \lambda \text{id})^k(\mathbf{v}) = \mathbf{0}$ then $(T - \lambda \text{id})^{k+1}(\mathbf{v}) = (T - \lambda \text{id})(T - \lambda \text{id})^k(\mathbf{v}) = \mathbf{0}$.

Lemma. Given a vector space $(V, \mathbb{C})$ with $T \in \mathcal{L}(V, V)$ and λ an eigenvalue of T , if there is some $h \in \mathbb{Z}^+$ for which $E_\lambda^h = E_\lambda^{h+1}$, then $E_\lambda^h = E_\lambda^{h+k}$ for all $k \in \mathbb{Z}^+$.

Proof. Suppose there is some $k \in \mathbb{Z}^+$ such that $\mathbf{v} \in E_\lambda^{h+k+1}$ but $\mathbf{v} \notin E_\lambda^{h+k}$. Then $(T - \lambda \text{id})^k(\mathbf{v}) \in E_\lambda^{h+1}$ but $(T - \lambda \text{id})^k(\mathbf{v}) \notin E_\lambda^h$, contradicting the claim $E_\lambda^h = E_\lambda^{h+1}$.

Definition. Given λ is an eigenvalue for $T \in \mathcal{L}(V, V)$, the **smallest** $h \in \mathbb{Z}^+$ for which $E_\lambda^h = E_\lambda^{h+1}$ is called the **height** of λ .

Generalised eigenspaces

From now on we shall assume V is a finite-dimensional vector space with $\dim(V) = n$, and λ is an eigenvalue of $T \in \mathcal{L}(V, V)$ with height h .

Lemma. We have $1 \leq h \leq n$ and $G_\lambda = E_\lambda^h$.

Proof. First note that $E_\lambda^k \leq V$ for all k , so $\dim(E_\lambda^h) \leq \dim(V)$ and $h \leq n$. Next, clearly $E_\lambda^h \subseteq G_\lambda$ by definition. But there can be no $\mathbf{v} \in G_\lambda$ that is not in E_λ^h , else there would be some k for which $\mathbf{v} \in E_\lambda^{h+k}$ in which case $E_\lambda^{h+k} \neq E_\lambda^h$, contradicting the fact h is the height of λ . So $G_\lambda = E_\lambda^h$.

Lemma. Writing $d_k = \dim(E_\lambda^k)$, we have

$$0 = d_0 < d_1 < d_2 < \cdots < d_{h-1} < d_h = d_{h+1} = \cdots \leq \dim(V).$$

Proof. Since E_λ^{h+1} is the first time that a generalised eigenspace equals the one before it in the sequence, all terms before it in the sequence must strictly increase in dimension.

Lemma. If $\mathbf{v} \in E_\lambda^{k+1} - E_\lambda^k$, then $(T - \lambda \text{id})(\mathbf{v}) \in E_\lambda^k - E_\lambda^{k-1}$.

Proof. Certainly we have that $(T - \lambda \text{id})(\mathbf{v}) \in E_\lambda^k$, but suppose also that $(T - \lambda \text{id})(\mathbf{v}) \in E_\lambda^{k-1}$. Then $\mathbf{0} = (T - \lambda \text{id})^{k-1}(T - \lambda \text{id})(\mathbf{v}) = (T - \lambda \text{id})^k(\mathbf{v})$, implying $\mathbf{v} \in E_\lambda^k$, which contradicts the fact $\mathbf{v} \in E_\lambda^{k+1} - E_\lambda^k$.

Corollary. Writing $\hat{E}_\lambda^{k-1}$ for a complementary subspace of E_λ^{k-1} (so that $E_\lambda^k = E_\lambda^{k-1} \oplus \hat{E}_\lambda^{k-1}$), we have $G_\lambda = E_\lambda^1 \oplus \hat{E}_\lambda^1 \oplus \hat{E}_\lambda^2 \oplus \cdots \oplus \hat{E}_\lambda^{h-1}$.

Jordan chains

The previous lemma shows that repeatedly applying $(T - \lambda \text{id})$ to a generalised eigenvector $\mathbf{v} \in E_\lambda^k - E_\lambda^{k-1}$ successively produces elements of smaller generalised eigenspaces down to some eigenvector $(T - \lambda \text{id})^{k-1}(\mathbf{v})$.

Writing $\mathbf{v}_i$ for an element of $E_\lambda^i - E_\lambda^{i-1}$ and $\mathbf{v}_1$ for an element of E_λ^1 , we can write this chain as

$$\mathbf{v}_k \xrightarrow{T - \lambda \text{id}} \mathbf{v}_{k-1} \xrightarrow{T - \lambda \text{id}} \dots \xrightarrow{T - \lambda \text{id}} \mathbf{v}_1 \xrightarrow{T - \lambda \text{id}} \mathbf{0}.$$

Definition. An ordered set of non-zero vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ for which $T(\mathbf{v}_1) = \lambda \mathbf{v}_1$ and $T(\mathbf{v}_i) = \lambda \mathbf{v}_i + \mathbf{v}_{i-1}$ for all $i > 1$ is called a **Jordan chain** of length k for the eigenvalue λ .

A Jordan chain is called **maximal** if it is not contained in a chain of greater length.

Lemma. The vectors in a Jordan chain are linearly independent.

Proof. Suppose $\alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0}$ for a Jordan chain $\mathbf{v}_1, \dots, \mathbf{v}_k$ of non-zero vectors, and scalars $\alpha_i \in \mathbb{C}$. Applying $(T - \lambda \text{id})^{k-1}$ to both sides implies $\alpha_k = 0$, then applying $(T - \lambda \text{id})^{k-2}$ to both sides implies that $\alpha_{k-1} = 0$, and so on down to $\alpha_1 = 0$. So the vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent.

Jordan blocks

Notice that for any Jordan chain $\mathbf{v}_1, \dots, \mathbf{v}_k$, since $T(\mathbf{v}_1) = \lambda \mathbf{v}_1$ and $T(\mathbf{v}_i) = \lambda \mathbf{v}_i + \mathbf{v}_{i-1}$ for all $i > 1$, the transformation matrix for T restricted to $\text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\})$ over the basis $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a $k \times k$ matrix of the form

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda & 1 & \dots & 0 & 0 \\ 0 & 0 & \lambda & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \lambda & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda \end{pmatrix}.$$

Definition. A matrix of the form above, with λ as every main diagonal entry, 1 as every superdiagonal entry (the $(i, i+1)$ th entry for each $1 \leq i < k$), and 0 as every other entry, is called a **Jordan block** for the eigenvalue λ of size k and notated as $J_k(\lambda)$.

We will now work towards proving the **Jordan normal form theorem**, which states that **every** linear transformation on a finite-dimensional vector space over $\mathbb{C}$ has a basis for which the corresponding transformation matrix is a direct sum of Jordan blocks (a **Jordan matrix**). This can be thought of as a generalisation of diagonalisation (allowing some superdiagonal entries to be 1).

Example

Example. Find a Jordan block J similar to the matrix $A = \begin{pmatrix} 2 & -9 \\ 1 & 8 \end{pmatrix}$.

Solution. The characteristic polynomial is $p_A(t) = \det(tI - A) = (t - 5)^2$. So 5 is an eigenvalue with algebraic multiplicity 2.

Now $E_5 = \ker(A - 5I) = \ker \begin{pmatrix} -3 & -9 \\ 1 & 3 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 3 \\ -1 \end{pmatrix} \right\}$, so the geometric multiplicity of 5 is 1, meaning A is not diagonalisable.

To build the corresponding Jordan block, we want to find a Jordan chain of length 2 within the generalised eigenspace G_5 . Noting that $(A - 5I)^2 = \mathbf{0}$, we see that any non-zero element of $\mathbb{C}^2$ that is not itself an eigenvector of A will belong to $E_5^2 - E_5^1$, so we can choose for example the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and

form the Jordan chain $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \xrightarrow{A-5I} \begin{pmatrix} -3 \\ 1 \end{pmatrix} \xrightarrow{A-5I} \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. This implies that we have $J = P^{-1}AP$ where $J = J_2(5) = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$ and $P = \begin{pmatrix} -3 & 1 \\ 1 & 0 \end{pmatrix}$.

Kernel-image decomposition

Lemma. Given a finite-dimensional vector space V with $\dim(V) = n$ and $T \in \mathcal{L}(V, V)$, we have $V = \ker(T^n) \oplus \operatorname{im}(T^n)$.

Proof. Suppose $\mathbf{v} \in \ker(T^n)$ and $\mathbf{v} \in \operatorname{im}(T^n)$. Then $T^n(\mathbf{v}) = \mathbf{0}$, and there is some $\mathbf{u} \in V$ for which $T^n(\mathbf{u}) = \mathbf{v}$. So $T^n(T^n(\mathbf{u})) = T^{2n}(\mathbf{u}) = \mathbf{0}$, implying $\mathbf{u} \in \ker(T^{2n})$. By a similar argument to that used for the map $T - \lambda \operatorname{id}$ earlier, we know that

$$\ker(T) \leq \ker(T^2) \leq \ker(T^3) \leq \cdots \leq \ker(T^n) \leq \cdots \leq V.$$

The dimensions of these kernels are weakly increasing along the chain, but cannot exceed dimension n , so in particular by the time we reach the $(n+1)$ th entry, we must have $\ker(T^{n+1}) = \ker(T^n)$. So in particular, $\ker(T^n) = \ker(T^{2n})$ and so $\mathbf{u} \in \ker(T^n)$, meaning $\mathbf{v} = T^n(\mathbf{u}) = \mathbf{0}$. Thus $\ker(T^n) \cap \operatorname{im}(T^n) = \{\mathbf{0}\}$, so their sum is a direct sum.

Now $\dim(\ker(T^n) \oplus \operatorname{im}(T^n)) = \dim(\ker(T^n)) + \dim(\operatorname{im}(T^n)) = \dim(V)$ by the rank-nullity theorem, so we must have that $V = \ker(T^n) \oplus \operatorname{im}(T^n)$.

Generalised eigenvector basis

Lemma. Given V is finite-dimensional with $T \in \mathcal{L}(V, V)$, there exists a **basis** for V composed of **generalised eigenvectors** of T .

Proof. Let $n = \dim(V)$. We proceed by induction n . In the base case where $n = 1$, note that all linear maps on a 1-dimensional vector space are of the form $T(\mathbf{v}) = \lambda \mathbf{v}$, such that every non-zero element of V is a λ -eigenvector.

Now suppose a basis of generalised eigenvectors exists for all vector spaces with dimension less than n . Let λ be an eigenvalue of T . From the previous lemma, we know that

$$V = \ker((T - \lambda \text{id})^n) \oplus \text{im}((T - \lambda \text{id})^n) = G_\lambda \oplus \text{im}((T - \lambda \text{id})^n).$$

Clearly if $\text{im}((T - \lambda \text{id})^n) = \{\mathbf{0}\}$, we are done. Otherwise, since $\dim(G_\lambda) > 0$, by the rank-nullity theorem we must have $\text{rank}((T - \lambda \text{id})^n) < n$ and so $\text{im}((T - \lambda \text{id})^n)$ has a basis of generalised eigenvectors by the inductive hypothesis. Extending this basis by a basis of G_λ creates a basis of eigenvectors for V .

Generalised eigenspace decomposition

Theorem. Suppose V is finite-dimensional and $T \in \mathcal{L}(V, V)$ has distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$. Then $V = G_{\lambda_1} \oplus G_{\lambda_2} \oplus \dots \oplus G_{\lambda_k}$.

Proof. We have shown that there exists a basis of V composed of generalised eigenvectors of T , so certainly $V = G_{\lambda_1} + G_{\lambda_2} + \dots + G_{\lambda_k}$.

Now suppose $\mathbf{v} \in G_\lambda$ and $\mathbf{v} \in G_\mu$ for some eigenvalues $\lambda \neq \mu$. Further suppose that $\mathbf{v} \neq \mathbf{0}$. Then $\mathbf{v} \in E_\lambda^\ell - E_\lambda^{\ell-1}$ and $\mathbf{v} \in E_\mu^m - E_\mu^{m-1}$ for some $\ell, m \in \mathbb{Z}^+$ (where in the case the height is 1, we interpret E_λ^0 as $\{\mathbf{0}\}$). So in particular, $\mathbf{w} = (T - \lambda \text{id})^{\ell-1}(\mathbf{v})$ is a λ -eigenvector.

Note that $(T - \lambda \text{id})$ and $(T - \mu \text{id})$ commute, since $(T - \lambda \text{id}) \circ (T - \mu \text{id}) = T \circ T - \lambda T - \mu T + \lambda \mu \text{id} = (T - \mu \text{id}) \circ (T - \lambda \text{id})$. So $(T - \mu \text{id})^m (T - \lambda \text{id})^{\ell-1}(\mathbf{v}) = (T - \lambda \text{id})^{\ell-1} (T - \mu \text{id})^m(\mathbf{v}) = \mathbf{0}$, implying $\mathbf{w} \in G_\mu$.

Now note that $(T - \mu \text{id})(\mathbf{w}) = T(\mathbf{w}) - \mu \mathbf{w} = (\lambda - \mu)\mathbf{w}$, so $(T - \mu \text{id})^m(\mathbf{w}) = (\lambda - \mu)^m(\mathbf{w}) = \mathbf{0}$. Since $\lambda \neq \mu$, we must have $\mathbf{w} = \mathbf{0}$, contradicting the fact $\mathbf{w}$ is a λ -eigenvector (thus non-zero). So we must have $G_\lambda \cap G_\mu = \{\mathbf{0}\}$.

Thus the sum of generalised eigenspaces must be a direct sum, that is, $V = G_{\lambda_1} \oplus G_{\lambda_2} \oplus \dots \oplus G_{\lambda_k}$.

Jordan chain decomposition

Lemma. Suppose V is finite-dimensional over scalar field $\mathbb{C}$ and $T \in \mathcal{L}(V, V)$. Then each **generalised eigenspace** of T has a basis composed of maximal **Jordan chains**. That is, for each distinct eigenvalue λ of T , writing $N = T - \lambda \text{id}$, there is a basis for G_λ of the form

$\{\mathbf{v}_1, N(\mathbf{v}_1), \dots, N^{m_1-1}(\mathbf{v}_1), \mathbf{v}_2, N(\mathbf{v}_2), \dots, N^{m_2-1}(\mathbf{v}_2), \dots, \mathbf{v}_k, N(\mathbf{v}_k), \dots, N^{m_k-1}(\mathbf{v}_k)\}$
for some $k \in \mathbb{Z}^+$, where $N^{m_i}(\mathbf{v}_i) = \mathbf{0}$ for some $m_i \in \mathbb{Z}^+$ for all $1 \leq i \leq k$.

Proof. Let $n = \dim(G_\lambda)$. We proceed by induction on n . In the base case where $n = 1$, clearly any non-zero element of $G_\lambda = E_\lambda$ will form a basis.

Now suppose a basis of Jordan chains exists for all generalised eigenspaces with dimension less than n . Consider the space $N(G_\lambda)$, which must have dimension less than n since if $\dim(N(G_\lambda)) = \dim(G_\lambda)$ we would have $N(G_\lambda) = G_\lambda$ implying λ is not an eigenvalue. By the inductive hypothesis, there must be a basis for $N(G_\lambda)$ of the form

$B_1 = \{\mathbf{w}_1, N(\mathbf{w}_1), \dots, N^{m_1-1}(\mathbf{w}_1), \mathbf{w}_2, \dots, N^{m_2-1}(\mathbf{w}_2), \dots, \mathbf{w}_k, \dots, N^{m_k-1}(\mathbf{w}_k)\}$
for some $\mathbf{w}_i \in N(G_\lambda)$ with $N^{m_i}(\mathbf{w}_i) = \mathbf{0}$ for all $1 \leq i \leq k$.

Note that since $N^{m_i-1}(\mathbf{w}_i) \in \ker(N)$ for each i , we can also take just these vectors and extend the set to a basis for $\ker(N)$ of the form

$B_2 = \{N^{m_1-1}(\mathbf{w}_1), \dots, N^{m_k-1}(\mathbf{w}_k), \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_\ell\}$ for some $\mathbf{u}_i \in \ker(N)$.

Jordan chain decomposition

Lemma. Suppose V is finite-dimensional over scalar field $\mathbb{C}$ and $T \in \mathcal{L}(V, V)$. Then each **generalised eigenspace** of T has a basis composed of maximal **Jordan chains**. That is, for each distinct eigenvalue λ of T , writing $N = T - \lambda \text{id}$, there is a basis for G_λ of the form

$$\{\mathbf{v}_1, N(\mathbf{v}_1), \dots, N^{m_1-1}(\mathbf{v}_1), \mathbf{v}_2, N(\mathbf{v}_2), \dots, N^{m_2-1}(\mathbf{v}_2), \dots, \mathbf{v}_k, N(\mathbf{v}_k), \dots, N^{m_k-1}(\mathbf{v}_k)\}$$

for some $k \in \mathbb{Z}^+$, where $N^{m_i}(\mathbf{v}_i) = \mathbf{0}$ for some $m_i \in \mathbb{Z}^+$ for all $1 \leq i \leq k$.

Proof (continued). For each $\mathbf{w}_i \in N(G_\lambda)$, choose $\mathbf{v}_i \in G_\lambda$ such that $N(\mathbf{v}_i) = \mathbf{w}_i$. Then we claim

$$B = \{\mathbf{v}_1, N(\mathbf{v}_1), \dots, N^{m_1}(\mathbf{v}_1), \dots, \mathbf{v}_k, N(\mathbf{v}_k), \dots, N^{m_k}(\mathbf{v}_k), \mathbf{u}_1, \dots, \mathbf{u}_\ell\}$$

is a basis for G_λ . To see that B is linearly independent, note that if $\alpha_1 \mathbf{v}_1 + \alpha_2 N(\mathbf{v}_1) + \dots + \alpha_j N^{m_k}(\mathbf{v}_k) + \beta_1 \mathbf{u}_1 + \dots + \beta_\ell \mathbf{u}_\ell = \mathbf{0}$, applying N to both sides gives a linear combination of the basis B_1 , so all but k of the α_i terms must be 0. Then the original expression becomes only a linear combination of the basis B_2 , so all other α_i and β_i terms must also be 0.

Also note that since $|B_1| = m_1 + m_2 + \dots + m_k$ and $|B_2| = k + \ell$,

$$|B| = (m_1+1) + (m_2+1) + \dots + (m_k+1) + \ell = \text{rank}(N) + \text{nullity}(N) = \dim(G_\lambda),$$

so B is indeed a basis for G_λ . This basis is composed of maximal Jordan chains (each $\mathbf{u}_i$ forms a chain of length 1), so the result follows by induction.

Jordan normal form

Theorem. Given that $(V, \mathbb{C})$ is a finite-dimensional vector space, every linear map $T \in \mathcal{L}(V, V)$ has a basis of V for which the corresponding transformation matrix is in Jordan normal form, that is, the direct sum of Jordan blocks.

Proof. This follows immediately from the previous lemmas. We know that V can be expressed as the direct sum of generalised eigenspaces G_λ for each distinct eigenvalue λ of T , and that each G_λ has a basis composed of distinct maximal Jordan chains. Using the union of these bases as the full basis for V , each maximal Jordan chain will correspond with a Jordan block, and so the full transformation matrix will be a direct sum of Jordan blocks.

Corollary. We have the following properties of the Jordan normal form for T , for each eigenvalue λ of T :

- The **sum of the sizes of the Jordan blocks** with diagonal λ (being the size of the basis for G_λ) is the **algebraic multiplicity** of λ .
- The **number of Jordan blocks** with diagonal λ (being the number of Jordan chains in the basis for G_λ) is the **geometric multiplicity** of λ .
- The **size of the largest Jordan block** with diagonal λ (being the size of the largest maximal Jordan chain in G_λ) is the **height** h of λ .

Example 1

Exercise. Given $A = \begin{pmatrix} 3 & -2 & 3 \\ 3 & -3 & 7 \\ 2 & -3 & 6 \end{pmatrix}$ and that $p_A(t) = (t - 2)^3$, find a

Jordan matrix J and invertible matrix P such that $J = P^{-1}AP$.

Solution. From the characteristic polynomial, we know that 2 is an eigenvalue with algebraic multiplicity 3. So A must be similar to $J_3(2)$ or $J_2(2) \oplus J_1(2)$ or $J_1(2) \oplus J_1(2) \oplus J_1(2)$.

$$\text{First find } E_2 = \ker \begin{pmatrix} 1 & -2 & 3 \\ 3 & -5 & 7 \\ 2 & -3 & 4 \end{pmatrix} = \ker \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}.$$

Since this has dimension 1, the eigenvalue 2 has geometric multiplicity 1 and so A must be similar to $J_3(2)$.

$$\text{Next find } E_2^2 = \ker((A - 2I)^2) = \ker \begin{pmatrix} 1 & -1 & 1 \\ 2 & -2 & 2 \\ 1 & -1 & 1 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Clearly $(A - 2I)^3 = \mathbf{0}$, so $E_2^3 = \mathbb{C}^3$. To start the Jordan chain, we can choose any element in $\mathbb{C}^3 - E_2^2$, for example $\mathbf{e}_1$. Then we get the chain

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \xrightarrow{A-2I} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \xrightarrow{A-2I} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \text{ so } P = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ 1 & 2 & 0 \end{pmatrix} \text{ and } J = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Example 2

Exercise. Given $A = \begin{pmatrix} 2 & 5 & -10 \\ 0 & 6 & -8 \\ 0 & 2 & -2 \end{pmatrix}$ and that $p_A(t) = (t - 2)^3$, find a

Jordan matrix J and invertible matrix P such that $J = P^{-1}AP$.

Solution. As in Example 1, A must be similar to $J_3(2)$ or $J_2(2) \oplus J_1(2)$ or $J_1(2) \oplus J_1(2) \oplus J_1(2)$.

$$\text{Here } E_2 = \ker \begin{pmatrix} 0 & 5 & -10 \\ 0 & 4 & -8 \\ 0 & 2 & -4 \end{pmatrix} = \ker \begin{pmatrix} 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \right\}.$$

Since this has dimension 2, the eigenvalue 2 has geometric multiplicity 2 and so A must be similar to $J_2(2) \oplus J_1(2)$.

For the $J_2(2)$ block, first note $(A - 2I)^2 = \mathbf{0}$, so $E_2^2 = \mathbb{C}^3$. To start the Jordan chain, we can choose any element in $\mathbb{C}^3 - E_2$, for example $\mathbf{e}_2$. Then

we get $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \xrightarrow{A-2I} \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix}$, and the $J_1(2)$ block corresponds with any element of E_2 that is linearly independent with these vectors, for example $\mathbf{e}_1$.

$$\text{So } P = \begin{pmatrix} 5 & 0 & 1 \\ 4 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix} \text{ and } J = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Example 3

Exercise. Given $A = \begin{pmatrix} 3 & -3 & 4 \\ 3 & -5 & 9 \\ 2 & -4 & 7 \end{pmatrix}$ and that $p_A(t) = (t-1)(t-2)^2$, find a

Jordan matrix J and invertible matrix P such that $J = P^{-1}AP$.

Solution. The eigenvalues are 1 and 2 with algebraic multiplicities 1 and 2 respectively, so A must be similar to $J_1(1) \oplus J_2(2)$ or $J_1(1) \oplus J_1(2) \oplus J_1(2)$.

Here $E_1 = \ker \begin{pmatrix} 2 & -3 & 4 \\ 3 & -6 & 9 \\ 2 & -4 & 6 \end{pmatrix} = \ker \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}$, so for

the $J_1(1)$ block we can choose an eigenvector from this span. Also,

$E_2 = \ker \begin{pmatrix} 1 & -3 & 4 \\ 3 & -7 & 9 \\ 2 & -4 & 5 \end{pmatrix} = \ker \begin{pmatrix} 1 & -3 & 4 \\ 0 & 2 & -3 \\ 0 & 0 & 0 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \right\}$, while

$E_2^2 = \ker((A - 2I)^2) = \ker \begin{pmatrix} 0 & 2 & -3 \\ 0 & 4 & -6 \\ 0 & 2 & -3 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} \right\}$, so A

must be similar to $J_1(1) \oplus J_2(2)$, and we can start the Jordan chain with $\mathbf{e}_1$

(say) to get $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \xrightarrow{A-2I} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$, so $P = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ 1 & 2 & 0 \end{pmatrix}$ and $J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.